
CYB813 - ADVANCED TOPICS IN CYBERSECURITY MANAGEMENT

Assessment 3: Scenario analysis and strategy report (Week 11)



Submitted by Room5_Group2

Name	Student ID
Poojit Bhaskar KASINA	24020655
Ashwini REGATTE	24020636
Dennis Kipkemei SANG	24021026
Betty Jepkorir MUGE	24020444

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Introduction

The actions taken by Silver Bank have shown how serious problems caused by weaknesses in structure, underdeveloped technology, and a lack of culture can combine to create one of the greatest threats we currently face, cybercrime. In a few years, Silver Bank was attacked electronically, and most incidents were deemed as micro level attacks, and the majority of those incidents were due to credential theft, phishing schemes, ATM fraud, and various other large ransomware threats that occurred against Silver Bank. These security breaches have shown all the different types of security holes that were within the security environment at Silver Bank and the poor risk management practices that have allowed for the security holes to happen. Here was an organisation that had had poor governance and an outdated governance framework relating to cybersecurity, used primarily legacy based technology, and had adopted a reactive method to dealing with its cybersecurity management, making it even more difficult for Silver Bank to monitor the changing nature of today's cyber threats.

Whereas Diamond Security was tasked with installing a standard cybersecurity protection program for the client. It can be observed that some positive results were attained during the process because the vulnerabilities that had been identified during the process, and even the ransomware attack on the bank. This demonstrated that the problems were beyond the technical measures that could be attained in the institution. Some of the key concepts that will be used to analyse the case are organisation structure, asset identification, threat intelligence, outsourcing, and adversarial/member simulations. An analysis of the governance and management issues in Silver Bank will produce a few recommendations to improve the security of the bank and build strong resilience.

Organisational and Leadership Considerations

Silver Bank's organisation was structured to set up major gaps in its cybersecurity posture. The lack of a Chief Information Security Officer (CISO) meant that all cybersecurity responsibilities fell on Tapia. This causing, both the IT and Operations Director, who was stretched to put the maximum operational demands. Since, there was no executive level leader devoted to security strategy, the risk oversight or policy formulation, cybersecurity was treated as a technical afterthought instead of an organisational priority as a strategy. These leadership gaps resulted in slow decision making. All potential threats were not visible as closeness to threat, and there was no one knows to handle or take accountability for security results.

The board directors at the governance level had a minimal cyber knowledge. With the basic awareness of cyber risk, there were numerous incidents occurred over several years at Silver Bank. Moreover, the board continued to deny to invest funding for cybersecurity. They believed that spending for IT was enough which shows a poor decision. With these limited sources, the need of upgrades being delayed, and vulnerabilities remained unresolved, like outdated SSL protocols, weak PKI infrastructure with weaker encryption, and a lack of proper monitoring systems. This caused the exposure of banks sensitive data. It is evident that governance failures were also evident with Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP) which are out of date. This lack of proper direction with regard to the 2020 ransomware attack caused employees being unprepared, and how to react and respond when the attack occurred. This indicated to develop clear incident response procedures or create an organisation where security was valued and gives more priority.

Silver Bank was always depended on outdated technology and ad hoc methods of improving and securing network. This included being unable to see future phishing scams due to no staff training and having not updated their security policies. The bank did not handle the incidents in a proper way. So therefore, it was difficult to respond effectively during a cyber attack which impacted entire bank operations. So, with the help of Diamond Security, the bank security was improved and during the ransomware incident, there was no planned security structure or framework in place. It was an ad hoc form of purple teaming. This clearly shows how the bank use heavy use of outside expertise. Slive Ban needs for better leadership, defined job functions and greater proactive management of cybersecurity.

Asset Identification and Risk Assessment

Silver Bank's way of doing asset identification and risk evaluation was kind of narrow, reactive, and mostly stuck on old inventories plus manual reviews. So the organisation couldn't really see what is happening in the most critical systems, user accounts, and the interdependent technologies between them. To show these flaws properly and not repeat stuff, the main assets, weaknesses, threat agents, and corresponding risk ratings are laid out in the table below. This setup brings forward the organisation's biggest exposure points, and ties each risk straight to proof coming from the case.

Asset	Vulnerability	Threat	Impact	Likelihood	Risk Rating	Evidence from case study
Staff credentials	No proper MFA, poor training	Phishing, credential theft Like passwords and expose of sensitive data	High	High	Critical	Repeated phishing incidents from 2015–2017
ATM Network Fraud	Legacy OS, weak monitoring	ATM malware inside the bank system	High	Medium	High	ATM fraud incident
Customer accounts	Outdated SSL, weak PKI include authentication and weaker encryption	Data theft, MITM	High	Medium	High	Outdated encryption protocols
Core banking system	Legacy infrastructure	Ransomware	Very High	Medium	Critical	2020 ransomware attack

Remote devices	No endpoint controls	Malware, lateral movement	Medium	High	High	Staff working remotely
Third party systems	No vendor risk assessment	Supply chain attack	High	Medium	High	Reliance on Diamond Security
BCP/DRP	Outdated, unclear	Operational disruption	High	High	Critical	Staff were not trained and did not know what to do and how to spot phishing attacks.

Dynamic Asset Mapping through Tools, Continuous Monitoring is a better alternative approach in the context of Silver Bank. Vulnerabilities Threat Modelling would yield a more reliable assessment of the relative levels of threat by modelling the exploits that an attacker could use against the resources, and therefore prioritise how those resources are going to be protected. Through adopting ISO 27001, as suggested by Diamond Security, a classification system of assets can be developed according to international standards. The use of AI for anomaly detection can also aid in detecting unusual activities in real time so that issues do not have the opportunity to develop. By utilising a Holistic Predictive Approach that combines these three aspects: Physical Assets, Human Assets, and Digital Assets, in conjunction with an Ongoing Risk Assessment, Silver Bank would be able to receive a Comprehensive Ongoing Risk Assessment that is Holistic in Nature to meet the ever changing and increasing nature of Cyber Attacks.

Intelligence and Decision Making

In the Silver Bank case, the role of threat intelligence was important, especially when a ransomware attack occurred against the bank in 2020. The FBI's notice about an impending global cyberattack is a perfect example of an external form of intelligence. This should have triggered immediate action on the part of Silver Bank, but this was not prepared to receive, analyse or escalate those types of alerts in any organised manner. On the other hand, Diamond Security used this intelligence as part of their operational response to help differentiate between the diversionary attacks on personal accounts and the actual threat to high value business accounts. This shows how effective use of intelligence can result in informed tactical decision making during an incident.

Ramírez was the primary individual responsible for internal threat intelligence collection (including information about system vulnerabilities, malware activity and unusual behaviours across the entire network) and acted previously as the intermediary of sorts when collecting this data. The use of operational intelligence allowed for the quick implementation of containment measures (for example, isolating infected machines) and prioritising the security of key business accounts; however, since so much reliance was placed on an external consultant (due to

Silver Bank not having dedicated internal resources/capacity) that this could not have been managed without a key person overseeing the process, such as a Chief Information Security Officer (CISO).

This case study describes how important it is for an organisation to continually monitor and assess structured data as well as have long term intelligence integration. Silver Bank did not follow existing frameworks like MITRE ATT&CK to help them identify how attackers would operate, and therefore, they could not correlate prior incidents to new ones, creating no situational awareness, making them a reactionary organisation regarding threats. To improve their resilience and lessen the effects of future attacks, they should establish better threat intelligence practices through real time data monitoring. Also, implementing a clear escalation workflow and strategically developing intelligence from Threat Intelligence programs into governance for making decisions.

Outsourcing

Silver Bank depended greatly upon the outsourcing of their cybersecurity through the employment of Diamond Security to provide them with rated and highly trained staff who were able to support the security process for them. The benefit that came out of employing such services for Silver Bank during the ransomware attack situation was enormous, considering the importance of such services in managing a situation like this one. Such things, among many others, prove that employing an external service provides a more efficient way of dealing with stressful situations.

Moreover, it illustrates that Silver Bank became too dependent on others for providing those services which they could not provide themselves (due to their inability), they made use of old technology and lacked a security expert. Outsourcing is regarded as a substitute for the governance structure within. In other words, no proper expectations/performance criteria were established regarding outsourcing between Silver Bank and the outsourcing vendor. No developed governance model existed to facilitate the relationship. Hence, all the security risks pointed out by Diamond Security (such as weakness in PKI/SSL) had remained unattended until they came to light.

Outsourcing requires internal leadership to help guide the process of outsourcing, clear definition of functions for those involved in outsourcing and continuous supervision of outsourcing. The experience of Silver Bank in outsourcing has revealed that outsourcing might lead to improvements in terms of security but cannot substitute governance, insufficient transparency regarding asset ownership or absence of security culture internally. Proper balance between external and internal resources is needed to achieve sustainability in the future.

Red/Blue/Purple Teaming

Red teaming refers to simulating an enemy's attacks in order to find vulnerabilities. For this case, Diamond Security's previous audits found out of date SSL protocols and weak PKI support, as well as gaps in monitoring, which are formal duties of pen test, in which a red team assesses systems for potentially exploitable vulnerabilities rather than adversarial intent. During their more in depth analysis of the ransomware incident, they found out that not only was there a planned attack on personal accounts as a diversion, but that the main targets of the actual

attack were the company's business accounts, reflecting how red team operations think through the eyes of the hacker.

In the incident that occurred in 2020 with Silver Bank, there was a need to detect and respond to attacks through the blue team(s); in this case, the actions taken by the ICT Staff at Silver Bank and through Diamond Security included isolating infected systems, safeguarding high value accounts and restoring services. However, because there were no currently deployable technical resources, and because there was not a sufficient amount of procedural resources that had been developed for the purposes of responding to and remediating incidents. Because of a lack of ongoing monitoring, the blue team security capability at Silver Bank was limited. Also, the employees who stated that they “did not know what to do” give a clear picture that there is a lack of maturity of the blue team function and a clear need for structured incident response processes to be in place.

The collaboration between Silver Bank and Diamond Security during the ransomware incident can be seen as a bit of a real example of how you might run a Purple Team process. In a way, Purple Teaming blends overall security posture, which gets stronger. The way both sides worked together was sort of through informal sharing of intelligence, doing joint analysis, and then coordinating the response. Purple Team Model is used like using a shared framework or frameworks such as MITRE ATT&CK, which helps Silver Bank lay out attack paths in a systematic manner. From there, they can spot where detection was missing and also figure out where to shore up the controls. If the silver bank keeps a broad approach with structured Red, Blue, and Purple Teaming together, the organisation tends to be more capable of preparing for and then surviving future incidents.

Recommendations

For the improvement of cybersecurity capacities at Silver Bank, several initiatives must be taken all together in the areas of governance, engineering and corporate culture. First, there is a need to introduce the role of Chief Information Security Officer (CISO) as an advisor of the highest rank for cybersecurity, threat management, and threat intelligence/incident response/security operations. The CISO will have to report to the Board of Directors, thus providing cybersecurity with high visibility within the corporation.

Secondly, the introduction of a formalized governance framework for cybersecurity will be required – for example ISO 27001 or NIST CSF, that will define processes to be followed while performing asset identification and classification, performing a comprehensive risk assessment, implementing controls, maintaining asset inventory (as dynamic as possible), and conducting periodic risk assessment for all important corporate assets, such as customer data storage, ATM network, authentication servers etc.

Thirdly, there is a need for Silver Bank to upgrade their technological framework with the assistance of upgraded legacy systems using advanced technologies, advanced encryption techniques, and endpoint protection for all devices. Furthermore, multi factor authentication (MFA), advanced management of public key infrastructure (PKI) and constant monitoring for all segments of the firm would also be required.

Fourthly, Silver Bank is mandated to upgrade its threat intelligence capability with the aid of processes that assist in receiving, analysing, and escalating external alerts. Some of the other elements include enhanced monitoring techniques using logs and anomalies detection among others in addition to MITRE ATT&CK2 frameworks.

Conclusion

From this case study, we have seen that they have had cybersecurity issues not only where there are technical weaknesses but wider organisational weaknesses as well. We can observe that due to weak governance leadership, unclear roles, and outdated technologies. The bank have susceptible to various forms of attacks, including phishing and ATM fraud, and ransomware

When incidents occurred, staff were unsure what was required of them to adequately respond to an incident. Silver Bank relied heavily on third parties for assistance with responding/mitigating a breach as opposed to relying on its own staff. Silver Bank needs better leadership (and therefore clearer communication), the use of up to date technology, as well as processes to manage risks, are all critical towards improving this environment. When Silver Bank starts taking a more proactive approach with organised roles in their operations. They will be better positioned to secure their critical systems and respond to future cyber threats in a better way.

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